

Math 421

Monday, December 1

Announcements

- Drop-In Hours
 - Wednesday 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Homework 10 due Friday (last HW!)

Mean Value Theorem

Theorem 29.3. If f is continuous on $[a, b]$ and differentiable on (a, b) , then there is a number $x_0 \in (a, b)$ such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}$$

Proof of MVT

Proof.

Constant Functions

Corollary 29.4. If f is defined on (a, b) and $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b) .

True/False: If $f'(x) = 0$ for all $x \in \text{dom}(f)$ then f is constant.

Proof of Corollary

Corollary 29.4. If f is defined on (a, b) and $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b) .

Proof.

Activity – Prove the Corollary

Corollary 29.5. If f and g are defined on (a, b) and $f'(x) = g'(x)$ for all $x \in (a, b)$, then there is some number c such that $f(x) = g(x) + c$ for all $x \in (a, b)$.

Proof.

Increasing & Decreasing

Def. Let f be a real-valued function defined on an interval I . If $x_1, x_2 \in I$ and $x_1 < x_2$ we say

1. f is **increasing** on I if $f(x_1) \leq f(x_2)$ and
2. f is **strictly increasing** on I if $f(x_1) < f(x_2)$

Examples:

Another Corollary of the MVT

Corollary 29.7. Let f be a differentiable function on an interval (a, b) . Then

1. f is increasing if $f'(x) \geq 0$ for all $x \in (a, b)$.
2. f is strictly increasing if $f'(x) > 0$ for all $x \in (a, b)$.

Q: How does this follow from the MVT?

Motivating Question & Activity

Q: Is the derivative of a differentiable function always continuous?

Activity: Let $f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Calculate, using the derivative rules and the fact that $(\sin x)' = \cos x$, $f'(a)$, $a \neq 0$.

Example

Let

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then

$$f'(x) = \begin{cases} -\cos \frac{1}{x} + 2x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

IVT for Derivatives

Theorem 29.8. Let f be a differentiable function on (a, b) . Then $f'(x)$ has the intermediate value property on (a, b) . That is:

If $a < x_1 < x_2 < b$, and if c lies between $f'(x_1)$ and $f'(x_2)$, there exists (at least one) x_0 in (x_1, x_2) such that $f'(x_0) = c$.

Activity – Prove the Theorem

Step 1: Define $g(x) = f(x) - cx$. Why does this make sense as a choice?

Step 2: Assume $f'(x_1) < c < f'(x_2)$. Show that $g(x)$ assumes its minimum at some point $x_0 \in (x_1, x_2)$. Why can't the minimum occur at the endpoints?

Step 3: Get that $g'(x_0) = 0$ and show that this gives the desired conclusion.